TOS DESIGN OPTIMIZATION: PLANT DIVERSITY SAMPLING

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1 BACKGROUND AND GOALS

The National Ecological Observatory Network (NEON) generates data and samples to advance understanding of ecological processes and inform management of US ecosystems. Multiple tiers of requirements provide traceable links between this high-level NEON mission statement and the protocols that direct data collection. The Terrestrial Observation System (TOS) science design for plant diversity ([RD03]) and the resulting study design, protocol ([RD03]), and data product (Plant Presence and Percent Cover, DP1.10058.001) are designed to enable status and trends in composition, abundance, and richness of plant species and function types at multiple scales.

1.1 Description of sample design and available data

NEON's plot-based sampling of plant species diversity satisfies the requirements that "NEON's measurement strategy will include biological invaders and diversity of key organismal groups" and "NEON measurements of plant diversity will include native and invasive plant species (Schimel et al. 2011)." Species are observed with the Carolina Vegetation Survey (Peet et al. 1998) multi-scale plot design that borrows from a sampling approach pioneered by Robert Whittaker (Shmida 1984, Stohlgren 1995). NEON samples with a 20 x 20m square plot with four 10m x 10m subplots containing nested 10m^2 and 1m^2 subplots (Figure 1). Specifically:

- The identity of each species (or the lowest taxonomic rank possible) according to naming conventions maintained by the US Department of Agriculture, Natural Resources Conservation Service PLANTS database (USDA, NRCS 2016) is recorded in each subplot - eight 1m^2, eight 10m^2, four 100m^2.
- Estimates of taxon-specific abundance is made with estimates of cover within the 1m^2 subplots.

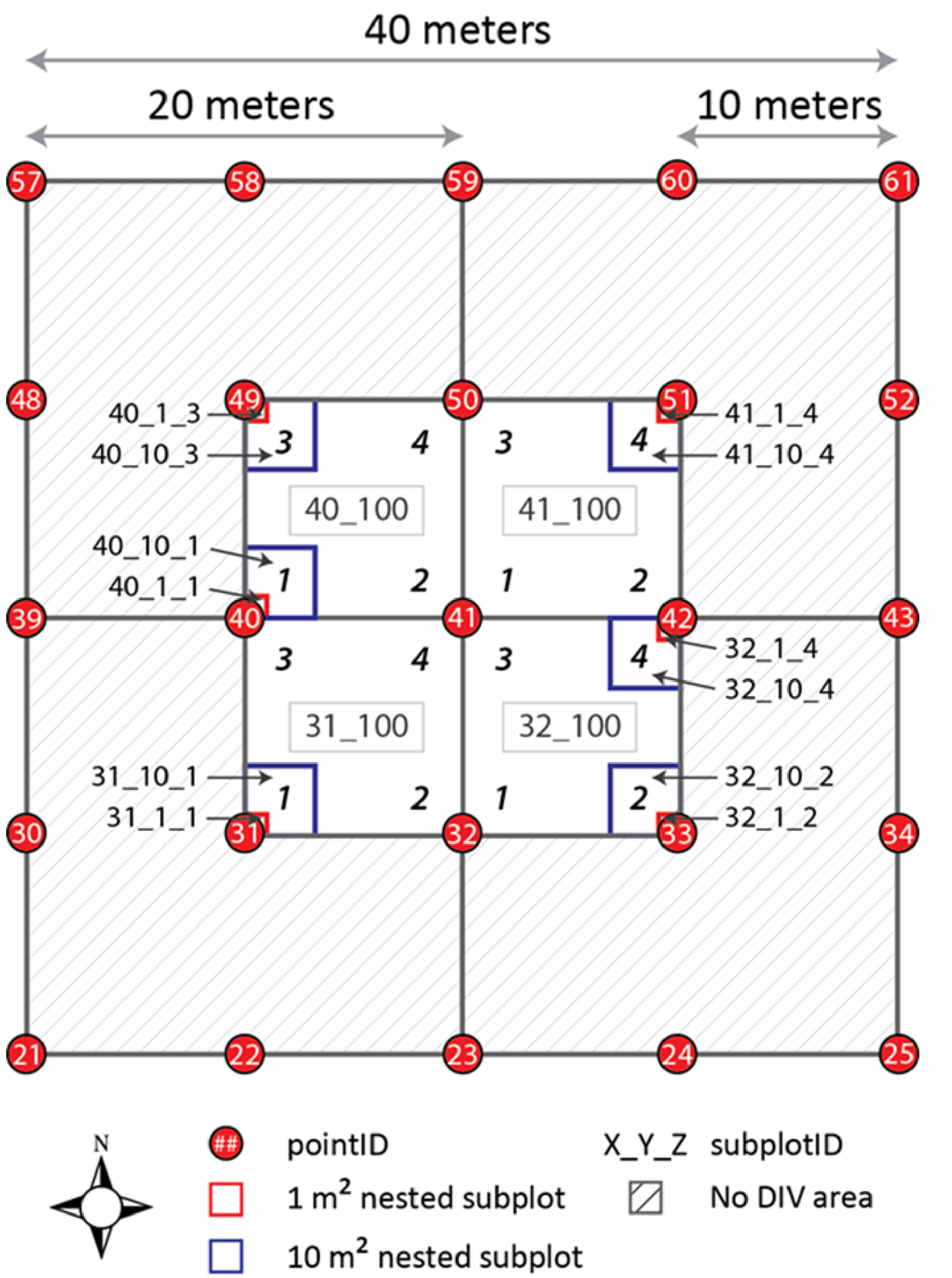

Figure 1: A diagram of the multiscale plot for observations of plant cover-abundance and occurrence data.

At each of 47 terrestrial NEON sites, the NEON spatial sampling design for organisms and soil directs observations to (Thorpe et al. 2016):

- Tower Base Plots: Randomly placed plots near the NEON flux tower designed to calibrate sensor-based measures of atmosphere and soil
- Distributed Base Plots: Plots to measure variability across larger landscapes according to a stratified-random design

The plant diversity protocol is implemented at three of the 20–30 Tower Base Plots. The majority of the plant species diversity sampling effort occurs at the Distributed Base Plots (Thorpe et al. 2016). Ecological literature (e.g., Stohlgren 2007), early input from the ecological community, and power analyses (Thompson 2012, Barnett et al. 2018) resulted in an initial sample size of 20–30 Distributed Base plots at each site.

The frequency and timing of sampling is guided by NEON requirements (Schimel et al. 2011). Implementing the protocol at each plot at least one time each year enables the quantification of annual rates of change. These sampling bouts occur during a one to two month period when the majority of species flower or possess other parts conducive to identification. This sampling window targets annual peaks in greenness as measured by MODIS, and incorporates input from local ecologists and NEON phenology observations (Meier et al. 2023). The few NEON sites characterized by distinct bimodal patterns in peak greenness populated disparate suites of apparent species are sampled twice annually. For example, the Sonoran desert (represented by the NEON site at the Santa Rita Experimental Range) gets rain in late winter and a monsoonal moisture pulse late in the summer. Plant species adapted to the particular intersection of temperature and the timing and amount of precipitation grow and flower and can be identified only during one of the two distinct events (Ogle and Reynolds 2004).

For the purpose of these analyses, data includes all available Distributed Base Plots from 46 NEON sites (D08 Dead Lake not included due to discontinued sampling, NEON 2025) downloaded directly from the application programming interface (API) and stacked with 'neonUtilities::stackByTable()' function on 03/01/2026.

1.2 Analytical Goals

Statistically rigorous analyses are needed to assess the capacity of NEON data to address Observatory goals and to guide iterative sampling design optimization efforts. Initial spatial and temporal sampling designs for the NEON Terrestrial Observation System (TOS) and Aquatic Observation System (AOS) were developed in collaboration with Technical Working Groups (TWGs) comprised of community experts, and were captured in Science Design documents (RD[01]). The initial designs relied on analysis of published datasets (where relevant), analysis of NEON prototype data collection efforts, and subject matter expertise.

As the NEON Observatory matures, it is critical for initial design assumptions to be tested with multiple years of data collected from NEON sites. Analyses must:

- Describe where the data support reduction of sampling effort without compromising NEON's ability to meet design requirements (e.g., target sites for reduction in sampling effort, spatial scales where replication is not necessary, etc.).

- Evaluate effects of design modifications on uncertainty and ability to detect changes for key response variables (e.g., effects of reduced spatial and/or temporal sampling effort).

Analysis and evaluation of the data provide a feedback loop that enables assessment of the designs relative to Observatory goals. Moreover, results of these analyses will allow the NEON to effectively prioritize sampling costs in the face of uncertain future funding levels.

Specification of analyses suitable for evaluating the efficacy of the NEON sampling designs required development of specific questions. Power analyses and sampling simulations were then used to determine the degree to which design modifications affected the ability of the data to address those questions. Questions and analytical approaches were developed in consultation with NEON Technical Working Groups comprised of subject matter experts selected from the ecological community.

The two primary and actionable questions of a 2018 effort for these data were:

1. Where is the variability in the data? Specifically, what proportion of the variance at each site exists at the year, bout, plot, and subplot scales? Understanding the components of variance provides a critical component of the actionable directions for modifying sampling size, subplot sampling intensity, and temporal sampling frequency.
2. Is it possible to detect meaningful interannual variation in key response variables? Understanding the power to detect change between years provides a direct indication of the sample sizes required to detect meaningful ecological change.

The variance components analysis supported the reduction in the number of nested 10m^2 and 1m^2 subplots within each plot from eight to six. Additionally, the 10m^2 and 100m^2 subplots at which species presence are recorded were reduced to every other year sampling. The 1m^2 subplots at each plot continue to be sampled annually.

The power analysis reduced the total number of plots sampled at a small number of sites. Only sites with three or more years of data were considered. An initial goal was to evaluate trend detection, but it was decided insufficient years of data were available.

With many more years of data available across all NEON terrestrial sites in 2026 (Appendix A), this effort revisits the question of detectability of interannual change in plant community composition and abundance. Specifically, we ask:

1. What level of spatial replication (number of plots) is required to reliably detect a specified relative change in plant species cover at a site?
2. What level of spatial replication is required for reliable estimation of multi-year directional trends in species cover?

Together, these analyses provide complementary bounds on the minimum sampling effort required to detect meaningful ecological change at short (interannual) and long (multi-year trend) temporal horizons.

2 RELATED DOCUMENTS AND ACRONYMS

2.1 Reference Documents

RD[01]	NEON.DOC.000001	NEON Observatory Design
RD[02]	NEON.DOC.002652	NEON Level 1, Level 2, and Level 3 Data Products Catalog
RD[03]	NEON.DOC.000912	TOS Science Design for Plant Diversity
RD[04]	NEON.DOC.014042	TOS Protocol and Procedure: Plant Diversity Sampling

2.2 Acronyms

Acronym	Definition
NEON	National Ecological Observatory Network
TOS	Terrestrial Observation System
TWG	Technical Working Group
NLCD	National Land Cover Database
GJAM	Generalized Joint Attribute Model
CA	Censored Abundance (GJAM likelihood for cover data)
MCMC	Markov Chain Monte Carlo

3 METHODS

We evaluated the capacity of NEON's terrestrial plant sampling design to detect ecological change in species- and community-level response variables using a hierarchical Bayesian modeling framework combined with simulation-based power and sensitivity analyses. The assessment comprises two complementary analyses:

1. We conducted a year-to-year change detection power analysis to quantify the spatial replication (number of randomly placed plots) required to detect a specified magnitude of interannual change in species cover, evaluated through a paired counterfactual design that controls for natural year-to-year variability.
2. We evaluated the spatial replication required for reliable estimation of multi-year directional trends, including the stability of trend estimates across sample sizes and the detectability of imposed trends of known magnitude.

Together, these analyses address the two primary temporal scales at which change detection is relevant to NEON's mission: abrupt interannual shifts and gradual directional trends.

3.1 Data and preprocessing

The observation unit for modeling is a species-specific cover value at the plot-year level. Within each plot-year, the six (or eight prior to 2019) 1m^2 subplot observations are averaged to produce a single cover value per species per plot per year. For example, at a site where 30 plots are sampled annually for 8 years, a species recorded on all plots would contribute 240 plot-year observations to the model. The full model input for a site is the matrix of all species × all plot-years, with cover values as entries.

Prior to model fitting, species occurring in fewer than three plots across the entire site record are aggregated into a composite "OTHER" category. This filter removes species for which spatial replication is insufficient to estimate covariance structure, while retaining them in the modeled community so that their aggregate contribution to total cover is represented (Clark et al. 2017). Species with zero variance or with no non-zero observations after data processing are removed, as they do not contribute information to model fitting. The "OTHER" category contributes to the model fit but is excluded from year-to-year change and trend analyses.

After the model fitting and posterior prediction steps, species with less than 3% mean cover per plot in the field are excluded from the year-to-year change detection and power analyses. A 20% relative change at 3% cover corresponds to 0.6 percentage points—below the 1% recording resolution of NEON's field protocol. This cover floor is computed from the raw field observations (mean of `percent cover` per species across all plot-years at the site) and is independent of the GJAM model predictions, which undergo compositional normalization that compresses cover values. It ensures that inference is focused on the portion of the community where meaningful change can be resolved, and the same filtered species set was used consistently across both the change detection and trend analyses.

3.2 Joint species distribution model

Both analyses share a common modeling framework. Plant species cover was modeled with the Generalized Joint Attribute Model (GJAM; Clark et al. 2017), a multivariate hierarchical Bayesian framework that jointly represents all species at a site while accounting for shared responses to environmental covariates and residual covariance among species. A single GJAM model was fit per site using all available plot-year observations. Both the year-to-year change detection and trend analyses draw on the posterior samples of this shared fit. For plot p in year t, the vector of observed percent cover values for all S species is modeled as:

$$\mathbf{y}_{p,t} \sim \text{CA}(\mu_{p,t}, \Sigma)$$

$$\mu_{p,t} = \mathbf{X}_{p,t}\mathbf{B}$$

where $\mathbf{y}_{p,t}$ is the S-vector of observed percent cover, $\mu_{p,t}$ is the latent mean cover vector, $\mathbf{X}_{p,t}$ is the design matrix of covariates, $\mathbf{B}$ is the $K \times S$ matrix of species-specific regression coefficients (equivalent to the transpose of the coefficient matrix in Clark et al. 2017), and Σ represents residual covariance among species not explained by the covariates, where K is the number of covariates.

The censored-abundance (CA) likelihood represents observed cover as a censored realization of an underlying multivariate normal latent variable, allowing exact zeros as well as continuous positive values.

Two model specifications were used depending on land cover heterogeneity at each site: $\mathbf{Y} \sim$ year $+$ nlcdClass at sites with multiple National Land Cover Database (NLCD) classes represented among sample plots, and $\mathbf{Y} \sim$ year at sites with a single class or where the multi-class specification failed to converge.

Year was treated as a categorical factor. The MCMC sampler was run for 5,000 iterations; 2,500 iterations were discarded as burn-in, yielding 2,500 retained posterior draws. For downstream posterior prediction and sensitivity analyses, we used $G = 1{,}000$ posterior draws sampled from the retained MCMC output for computational efficiency.

3.2.1 Posterior prediction and dimension reduction

Joint modeling of plant species abundance data from NEON sites frequently involves hundreds of species observed across tens of plots, producing covariance matrices that are high-dimensional relative to the available sample size. For example, a southeastern forest site with 400 plant species observed across 30 plots per year requires estimating the 400×400 residual covariance matrix Σ from approximately 300 plot-year observations. This implies estimating on the order of $S(S + 1)/2 \approx 80{,}000$ covariance parameters from far fewer independent observations. Under these conditions, the covariance matrix is not fully identifiable: the sample covariance is rank-deficient, and parameter estimates are unstable without additional structure. This limitation arises from the dimensionality of the problem and is exacerbated by the presence of many rare species with sparse observations.

GJAM addresses this challenge through a Bayesian dimension reduction framework (Taylor-Rodriguez et al. 2017; Clark et al. 2017). Rather than estimating Σ directly, the model represents the S-dimensional species response in a lower-dimensional latent space. Species are assigned to N_c clusters via a truncated Dirichlet process, and within each cluster, an r-dimensional factor loading matrix **Z** captures shared covariance. The implied covariance is:

$$\Sigma = \mathbf{A}\mathbf{A}' + \sigma_\varepsilon^2\mathbf{I}$$

where $\mathbf{A} = \mathbf{Q}(\mathbf{k})\mathbf{Z}$ is the $S \times r$ loading matrix that maps the reduced factor structure back to the full species space via cluster assignments **k**, and σ_ε^2 is a scalar residual variance. This factorization reduces the effective number of covariance parameters from $S(S+1)/2$ to $(N_c \times r) + 1$, enabling stable estimation even when S greatly exceeds the number of observations n.

GJAM invokes dimension reduction when the number of species is large relative to the number of observations (Taylor-Rodriguez et al. 2017). This was the case at all 46 NEON sites included in the analysis. Species counts ranged from 39 to 406 while plot-year observations ranged from approximately 100 to 400. Because NEON data triggered GJAM dimension overrides, we set reduction explicitly with a truncation level of $N_c = 20$ clusters and $r = 5$ factors across all sites to ensure consistent parameterization and comparability of posterior predictions across the Observatory. Posterior predictive draws were reconstructed from the stored MCMC chains for **B**, **k**, **Z**, and σ_ε^2 using the reduced-rank generative model:

$$\mathbf{y}_i^{(g)} = \mathbf{x}_i\mathbf{B}^{(g)} + \mathbf{w}_i^{(g)}\mathbf{A}^{(g)\prime} + \varepsilon_i^{(g)}$$

where $\mathbf{x}_i\mathbf{B}^{(g)} = \mu_i^{(g)}$ is the latent mean response defined above, $\mathbf{B}^{(g)}$ is the $K \times S$ matrix of species-specific regression coefficients for posterior draw g, $\mathbf{A}^{(g)}$ is the $S \times r$ loading matrix reconstructed from cluster assignments $\mathbf{k}^{(g)}$ and factor loadings $\mathbf{Z}^{(g)}$ at draw g, $\mathbf{w}_i^{(g)}$ is an r-dimensional latent factor score with $\mathbf{w}_i^{(g)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r)$, and $\varepsilon_i^{(g)}$ is species-level residual error with $\varepsilon_i^{(g)} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^{2(g)}\mathbf{I})$. Superscript

(g) indexes the posterior draw used for prediction ($g = 1, \ldots, G$, with $G = 1{,}000$ in the sensitivity analyses). For each draw, the parameters $\mathbf{B}^{(g)}$, $\mathbf{A}^{(g)}$, and $\sigma_\varepsilon^{2(g)}$ are fixed, while $\mathbf{w}_i^{(g)}$ and $\varepsilon_i^{(g)}$ are drawn independently, yielding posterior predictive realizations that preserve the covariance structure implied by the reduced-rank model without explicitly forming the full $S \times S$ covariance matrix.

The analysis propagates key sources of uncertainty through the posterior predictive distribution. Parameter uncertainty is captured through variation across MCMC draws, while residual variability—representing ecological and measurement variation not explained by covariates—is captured through the stochastic components $\mathbf{w}$ and ε. Observation uncertainty (field measurement error) is not modeled explicitly but is implicitly absorbed into the residual variance estimates.

3.3 Analytical framework

Both analyses use a common modeling and prediction framework to evaluate the detectability of ecological change. Posterior predictive draws from a single site-level model provide a representation of expected species abundance under observed species abundance and occurrence and environmental and spatial variability. These predictions are then combined with controlled perturbations and plot-level subsampling to quantify how sampling intensity influences inference.

The two analyses differ only in the temporal structure of the signal being evaluated. The year-to-year change detection analysis focuses on discrete changes between paired states, while the trend analysis evaluates directional change accumulated over multiple years. In both cases, inference is based on the behavior of posterior predictive summaries under repeated subsampling of plots, allowing direct comparison of how sampling design affects the ability to detect and estimate changes in plant species abundance.

3.3.1 Definition of perturbation, detection, and stability thresholds

Both analyses rely on three key elements: a specified perturbation magnitude, a detection criterion, and (for trend analysis) a stability criterion. Perturbations define the magnitude of change to be detected, detection criteria define when a change is considered statistically distinguishable, and stability criteria define when estimated trends are robust to spatial sampling.

For year-to-year change detection, perturbations were applied as proportional shifts in species cover, with a primary focus on a 20% relative change consistent with the 2018 NEON design analysis (Barnett et al. 2018). Additional effect sizes (15–30%) were evaluated to characterize sensitivity across a range of plausible ecological changes. The appendix results confirm that 20% represents an intermediate effect size where detectability varies meaningfully with sampling effort (Appendix A3). In the trend analysis, a 20% cumulative change over the observation window was used to define a comparable threshold for directional change.

In both analyses, detection required the posterior probability that an estimated change exceeded a specified threshold to reach a criterion of 0.80. For each species, this criterion was evaluated across posterior draws within each Monte Carlo replicate, and a species was classified as detectable when the criterion was satisfied in at least 80% of replicates. Community-level detection summarizes the percentage of species that were individually detectable at each sample size. This threshold balances sensitivity to ecological change with robustness to spatial variability and is consistent with common practice in detectability analyses.

For the trend analysis, a complementary stability criterion was defined based on the coefficient of variation (CV) of slope estimates across spatial subsamples. A threshold of CV ≤ 0.50 was used to indicate that variation in plot selection does not qualitatively alter the inferred trend, corresponding to a regime in which variability in slope estimates is small relative to their magnitude.

3.4 Year-to-year change detection (counterfactual analysis)

The objective of the counterfactual analysis is to evaluate the extent to which changes in species abundance can be detected given the spatial variability observed within NEON sites. Because plant communities are spatially heterogeneous, the detectability of change depends not only on the magnitude of the signal, but also on the number and configuration of plots used to estimate site-level means.

To address this, we used posterior predictive draws from a single site-level model to generate paired baseline and perturbed realizations of species cover from the same underlying year. The perturbation represents a controlled, biologically meaningful change applied to the latent mean structure, allowing direct comparison of predicted site-level responses under identical underlying variability.

For each site, detection was evaluated as a function of the number of plots sampled per year (K). Subsampling plots introduces variability analogous to field sampling, allowing the analysis to quantify how sampling intensity influences the probability of detecting a specified change. The result is a set of detection curves that describe the relationship between sampling effort and the probability of detecting a change of magnitude δ.

This framework isolates the role of spatial sampling in determining detectability. Because the baseline and perturbed predictions are derived from the same posterior draws, differences between them reflect only the imposed change and the variability introduced by subsampling, rather than differences in model fit or parameter uncertainty.

This analysis was motivated by earlier approaches that evaluated change between consecutive years, but extends that framework by removing the confounding influence of interannual ecological variability. Rather than comparing observations from different years, both baseline and perturbed predictions are generated from the same underlying data and posterior parameter draws, with independent realizations of observation-level variability (**w** and ε). This counterfactual design isolates statistical detectability from ecological variability among years, providing a controlled evaluation for monitoring performance under realistic covariance structure.

Let $\mu_{s,p}^{(g)}$ denote the posterior predictive cover for species s at plot p for MCMC draw g.

The perturbation was applied directly to the posterior predictive draws as a per-species additive shift proportional to each species' mean raw field cover:

$$\mu_{s,p}^{\text{pert},(g)} = \mu_{s,p}^{(g)} + \delta \cdot \bar{c}_s$$

where $\mu_{s,p}^{(g)}$ is the posterior predictive cover for species s at plot p for draw g, $\mu_{s,p}^{\text{pert},(g)}$ is the corresponding perturbed value, $\bar{c}_s$ is the mean percent cover of species s computed from raw field observations across all plot-years at the site, and δ is the relative effect size. A species at 10% mean cover with $\delta = 0.20$ receives a +2 percentage point perturbation; a species at 60% mean cover receives +12 percentage points. This per-species scaling ensures every species faces the same proportional challenge regardless of baseline cover level. We tested effect sizes of $\delta \in \{0.15, 0.20, 0.25, 0.30\}$, corresponding to 15–30% relative changes in cover.

For each site and year, the baseline and perturbed prediction sets were subsampled to represent alternative levels of spatial replication. All plots sampled in a given year were eligible for subsampling. No requirement was imposed that plots be sampled in consecutive years, as the paired counterfactual design generates both prediction sets from the same year's data. Subsamples consisted of K plots drawn without replacement, with $K \in \{30, 25, 20, 15, 10, 8, 6, 5\}$, and each subsampling was repeated 50 times.

For each Monte Carlo replicate, the site-level mean cover for species s was computed for both the baseline and perturbed prediction sets by averaging across the K sampled plots within each posterior draw g:

$$\bar{\mu}_{s,K}^{(g)} = \frac{1}{K}\sum_{p=1}^{K} \mu_{s,p}^{(g)}, \quad \bar{\mu}_{s,K}^{\text{pert},(g)} = \frac{1}{K}\sum_{p=1}^{K} \mu_{s,p}^{\text{pert},(g)}$$

where $\mu_{s,p}^{(g)}$ and $\mu_{s,p}^{\text{pert},(g)}$ denote the baseline and perturbed posterior predictive cover values, respectively.

We evaluated detection using a per-species absolute difference criterion. For each species s, detection probability was defined as:

$$P\left(\left|\bar{\mu}_{s,K}^{\text{pert},(g)} - \bar{\mu}_{s,K}^{\text{base},(g)}\right| \geq \tau_s\right) \geq 0.80$$

where $\bar{\mu}_{s,K}^{\text{base},(g)}$ and $\bar{\mu}_{s,K}^{\text{pert},(g)}$ are the site-level mean cover values computed from the baseline and perturbed prediction sets, respectively, for posterior draw g, and τ_s is the species-specific detection threshold. The probability is computed as the fraction of posterior draws g for which the absolute difference exceeds τ_s. The ratio of imposed effect to detection threshold is constant across species (2:1 at $\delta = 0.20$), providing equivalent statistical difficulty regardless of cover level. This ensures that detectability comparisons are not driven by differences in species abundance scale.

A species was classified as detectable at sample size K and effect δ when this criterion was satisfied in at least 80% of the 50 Monte Carlo replicates, corresponding to a detection probability of at least 0.80 under repeated subsampling. The community-level metric reports the percentage of species (above the 3% cover floor) that are individually detectable at each K. This definition evaluates whether an imposed change is distinguishable from background variability in the posterior predictive distribution.

3.5 Multi-year trend analysis

The trend analysis addresses a complementary question to the counterfactual framework: whether directional change in species abundance can be reliably estimated through time. The focus shifts from detecting a difference between two states to estimating the rate of change across multiple years.

As with the year-to-year change detection analysis, inference is based on posterior predictive draws from a single site-level model. For each species, predicted cover values are generated for all plot–year combinations, preserving the spatial and temporal structure of the observed data. These predictions are then used to estimate annual site-level means under varying levels of spatial replication, and form the basis for both observed trend detection and imposed trend analyses described below.

To evaluate the influence of sampling intensity, plots are subsampled independently within each year, and site-level mean cover is computed for each posterior draw. A linear trend is then estimated across years for each species, yielding a posterior distribution of slope parameters that reflects both model uncertainty and the variability introduced by spatial subsampling.

Two complementary properties of the estimated trend are evaluated. Detection assesses whether the magnitude of the estimated slope exceeds a specified threshold, analogous to the threshold-based detection used in the counterfactual analysis. Stability evaluates the sensitivity of the estimated slope to the selection of plots, quantified as the coefficient of variation of slope estimates across repeated subsamples at each sample size. Together, these metrics describe both the ability to detect directional change and the reliability of the estimated rate of change.

As in the counterfactual analysis, the objective is not to optimize model fit, but to quantify how spatial sampling constrains inference about ecological change. This question was identified as a priority during the 2018 NEON design optimization effort (Barnett et al. 2018) but could not be addressed at that time due to limited temporal replication. With 8–12 years of data now available at most NEON sites, the question is now tractable.

3.5.1 Moving-window framework

Trend detection was evaluated using a moving-window approach (Urquhart et al. 1998). Rather than fitting a single trend line through the entire time series, we applied overlapping windows of fixed width W that slide across the available years. This moving-window framework enables assessment of trends across multiple time periods and durations, while accommodating variation in the rate and magnitude of change.

For each site, we evaluated window widths of $W \in \{5, 10\}$ years. Within each window, a trend was quantified as the ordinary least-squares slope of species cover against year, computed from posterior predictive site-year means derived from plot-level predictions. This yields a posterior distribution of slope estimates for each species, window position, and sample size.

3.5.2 Observed trend detection

The observed trend detectability analysis evaluates whether trends already present in the posterior predictions are detectable as sample size changes. For each species, window position, posterior draw, and sample size K, the site-level mean cover was computed by averaging across the K sampled plots within each year:

$$\bar{\mu}^{(g)}_{s,t,K} = \frac{1}{K}\sum_{p=1}^{K}\mu^{(g)}_{s,p,t}$$

where $\mu^{(g)}_{s,p,t}$ is the posterior predictive cover for species s at plot p in year t for MCMC draw g. A temporal trend was then estimated for each species and posterior draw as the ordinary least-squares slope of $\bar{\mu}^{(g)}_{s,t,K}$ against centered year:

$$\gamma^{(g)}_{s,K} = \frac{\sum_t (t-\bar{t})\bar{\mu}^{(g)}_{s,t,K}}{\sum_t (t-\bar{t})^2}$$

A species was classified as detectable when the posterior probability that the absolute slope exceeded the threshold δ/T was at least 0.80:

$$P\left(|\gamma^{(g)}_{s,K}| \geq \frac{\delta}{T}\right) \geq 0.80$$

where $\gamma^{(g)}_{s,K}$ is the estimated trend slope for species s at sample size K, $\delta = 0.20$ corresponds to a cumulative 20% change over the window, and $T = t_{\text{max}} - t_{\text{min}}$ is the duration of the observation window. This definition evaluates whether the estimated rate of change is distinguishable from zero given spatial and temporal variability.

Plots were subsampled independently within each year, rather than requiring consistent plot identity across time. This accommodates variation in sampling effort among years (including reduced sampling in 2020) and reflects realistic monitoring conditions.

The evaluated sample sizes (K) span the range of sampling effort observed across sites, with a maximum target of $K = 30$. At sites with fewer available plots, the maximum evaluated K corresponds to the number of plots available. Summaries at the maximum sample size are based on the highest available K for each site, which may be less than 30.

3.5.3 Imposed trend detection

The imposed trend detection analysis evaluates the ability to detect an imposed trend of known magnitude and assess the number of plots needed at each site to detect that trend. This provides a prospective analysis analogous to the counterfactual change detection analysis, but for gradual directional trends rather than discrete changes at a single time point. In contrast to the year-to-year analysis, which evaluates a range of effect sizes, the trend analysis focuses on a 20% cumulative change over the observation window. This provides a consistent and interpretable benchmark for directional change while limiting computational burden associated with evaluating multiple trend scenarios.

For each window, an additive linear trend was imposed on the posterior predictive means before estimating slopes:

$$\mu_{s,p,t}^{\text{trend},(g)} = \mu_{s,p,t}^{(g)} + \frac{\delta}{T}(t - t_0)$$

where $\mu_{s,p,t}^{(g)}$ is the original posterior predictive cover, $\mu_{s,p,t}^{\text{trend},(g)}$ is the prediction after imposing the trend, t_0 is the first year of the window, δ is the imposed cumulative change, and T is the calendar time span of the window. Slopes $\gamma_{s,K}^{(g)}$ were then estimated from the trend-modified predictions using the same centered-year regression described above. Detection was assessed using the same posterior probability criterion as in the observed trend analysis.

3.5.4 Definition of N^* for trend detection

For each site, species, and window width, we defined N^* as the smallest number of plots K for which the trend detection criterion was met. As with the counterfactual change detection analysis, N^* values were computed separately for each species and then summarized at the community level.

3.5.5 Slope estimation stability

The detection-based analyses evaluate whether an estimated slope exceeds a fixed threshold. A complementary question is whether the slope estimate itself is stable, whether it would yield a similar value if a different set of plots were selected. A slope estimate that varies substantially depending on which plots are sampled provides a weak basis for inference, regardless of whether it exceeds a detection threshold.

To evaluate this, we computed the coefficient of variation (CV) of the slope estimate across Monte Carlo replicates at each sample size K. For each species s, window position, and sample size, each of the $R = 50$ Monte Carlo replicates yields a mean slope (averaged across posterior draws):

$$\bar{\gamma}_{s,K,m} = \frac{1}{G} \sum_{g=1}^{G} \gamma_{s,K,m}^{(g)}$$

where $\gamma_{s,K,m}^{(g)}$ is the slope estimate for species s at sample size K in Monte Carlo replicate m for posterior draw g, and G is the number of posterior draws.

The coefficient of variation across replicates is then:

$$\text{CV}_s(K) = \frac{\text{SD}\left(\bar{\gamma}_{s,K,1}, \dots, \bar{\gamma}_{s,K,R}\right)}{\left|\text{mean}\left(\bar{\gamma}_{s,K,1}, \dots, \bar{\gamma}_{s,K,R}\right)\right|}$$

At low K, slope estimates vary widely depending on which plots are selected (high CV); as K increases, estimates converge (low CV). A CV below 0.50 indicates that the standard deviation of the slope estimate is less than half its magnitude, such that variation in plot selection does not qualitatively change the inferred direction and strength of the trend. This threshold provides a practical criterion for inference, ensuring that variability in slope estimates does not overwhelm the estimated trend signal while remaining interpretable across species and sites.

This stability analysis is complementary to the detection-based analyses. Adding spatially heterogeneous plots to annual site means can increase year-to-year variability, making threshold-based detection more difficult at higher K—an effect that reflects real ecological heterogeneity but complicates interpretation of detection curves. In contrast, the CV metric evaluates the precision of the slope estimate itself and decreases monotonically with K at nearly all sites.

For each species and site, we defined K_{CV50} as the smallest sample size at which $\text{CV}_s(K) \leq 0.50$, representing the point at which slope estimates become stable with respect to plot selection.

3.6 Scope and relationship between analyses

The year-to-year change detection and trend analyses address ecological change at different temporal scales. The year-to-year change detection analysis quantifies the spatial replication required to detect a specified relative change at a single time point, providing insights on interannual shifts in plant species. The trend analysis evaluates the number of plots required for reliable trend estimation over multi-year windows, providing guidance relevant to tracking gradual directional changes. The slope stability analysis from the trend framework directly parallels the year-to-year change detection curves—both ask how the quality of inference improves with additional plots. Where both analyses converge on similar sample size requirements, the recommendation is strengthened.

4 RESULTS

We first summarize site-level data characteristics, including the number of years of observation and the number of species retained after applying the cover filter. We then present detailed results for a small number of representative sites to illustrate the behavior of each analysis, followed by summary tables that synthesize results across all completed sites.

4.0.1 Site characteristics

The number of years of observation ranged from 2013 to 2024, with most sites having at least eight years of data. This expanded temporal record enables evaluation of both year-to-year change detection and multi-year trend estimation.

The number of species observed across all plots and years ranged from 70 at an agricultural site in Colorado to 823 in Georgia. The number of species included in the GJAM model ranged from 39 to 406 per site. After applying the 3% mean cover threshold, the number of species retained for inference ranged from 4 to 122 per site (Table 1).

These differences reflect substantial variation in species richness and community structure across the Observatory, which in turn influences detectability and trend estimation.

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Table 1: Summary of site characteristics, including years of observation and species counts for observed, modeled, and inference subsets.

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species (≥3% cover)
D01	Bartlett Experimental Forest	BART	11	2014-2024	95	44	7
D01	Harvard Forest & Quabbin Watershed	HARV	11	2014-2024	322	145	47
D02	Blandy Experimental Farm	BLAN	10	2015-2024	292	149	30
D02	Smithsonian Conservation Biology Institute	SCBI	11	2014-2024	406	226	58
D02	Smithsonian Environmental Research Center	SERC	9	2015-2024	285	171	35
D03	Disney Wilderness Preserve	DSNY	10	2014-2024	612	304	92
D03	Ordway-Swisher Biological Station	OSBS	11	2013-2024	512	212	54
D03	The Jones Center At Ichauway	JERC	10	2014-2024	823	406	122
D04	Guánica Forest	GUAN	10	2015-2024	229	121	77
D04	Lajas Experimental Station	LAJA	9	2016-2024	226	108	83
D05	Steigerwaldt-Chequamegon	STEI	10	2015-2024	418	190	71
D05	Treehaven	TREE	10	2015-2024	411	191	49
D05	University of Notre Dame Environmental Research Center	UNDE	11	2014-2024	393	198	44
D06	Konza Prairie Agroecosystem	KONA	8	2017-2024	113	46	9
D06	Konza Prairie Biological Station	KONZ	9	2016-2024	355	195	52
D06	University of Kansas Field Station	UKFS	9	2016-2024	383	207	68
D07	Great Smoky Mountains National Park	GRSM	5	2015-2024	259	126	47

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species (≥3% cover)
D07	Mountain Lake Biological Station	MLBS	8	2017-2024	217	124	48
D07	Oak Ridge	ORNL	11	2014-2024	370	186	64
D08	Lenoir Landing	LENO	8	2016-2024	289	161	33
D08	Talladega National Forest	TALL	11	2014-2024	452	244	64
D09	Chase Lake National Wildlife Refuge	WOOD	11	2014-2024	282	127	45
D09	Dakota Coteau Field Site	DCFS	8	2017-2024	264	143	36
D09	Northern Great Plains Research Laboratory	NOGP	9	2016-2024	257	125	49
D10	Central Plains Experimental Range	CPER	12	2013-2024	206	122	22
D10	North Sterling	STER	11	2014-2024	70	39	16
D10	Rocky Mountains	RMNP	8	2017-2024	215	98	19
D11	Lyndon B. Johnson National Grassland	CLBJ	7	2016-2024	491	249	63
D11	Marvin Klemme Range Research Station	OAES	9	2015-2024	397	224	47
D12	Yellowstone National Park	YELL	6	2018-2024	421	185	29
D13	Moab	MOAB	8	2015-2024	174	95	4
D13	Niwot Ridge	NIWO	10	2015-2024	247	147	30
D14	Jornada Experimental Range	JORN	9	2015-2024	171	91	5
D14	Santa Rita Experimental Range	SRER	9	2016-2024	441	271	38
D15	Onaqui	ONAQ	10	2014-2024	181	90	9
D16	Abby Road	ABBY	8	2017-2024	247	129	43

Domain	Site	Site Code	Years	Year Range	Species (observed)	Species (modeled)	Species (≥3% cover)
D16	Wind River Experimental Forest	WREF	7	2018-2024	123	60	23
D17	Lower Teakettle	TEAK	5	2019-2024	199	70	29
D17	San Joaquin Experimental Range	SJER	8	2017-2024	252	130	40
D17	Soaproot Saddle	SOAP	6	2018-2024	317	131	46
D18	Toolik Field Station	TOOL	8	2017-2024	188	105	19
D18	Utqiaġvik	BARR	7	2017-2024	90	56	10
D19	Caribou-Poker Creeks Research Watershed	BONA	8	2017-2024	84	46	26
D19	Delta Junction	DEJU	9	2016-2024	182	98	24
D19	Healy	HEAL	10	2015-2024	136	78	36
D20	Pu'u Maka'ala Natural Area Reserve	PUUM	7	2018-2024	122	61	20

\endgroup

4.1 Year-to-year change detection (counterfactual analysis)

4.1.1 Example sites

We illustrate the behavior of the counterfactual change detection analysis with three representative sites: ABBY, GRSM, and BONA (Figure 2), spanning a range of community composition and spatial heterogeneity.

At all three sites, detection increased with sample size at the focal effect size of $\delta = 0.20$, but the rate and shape of that increase differed among sites. At ABBY, detection rose steadily with increasing K, approaching high detection probabilities at larger sample sizes. GRSM exhibited more rapid gains at smaller sample sizes but a more gradual approach to the detection threshold. At BONA, detection increased with K but did not reach the 80% threshold within the evaluated range, indicating that the imposed change could not be reliably detected even at maximum sampling effort.

At smaller effect sizes ($\delta = 0.15$), detection remained low across all sample sizes, indicating that changes below 20% are not reliably detectable under observed spatial variability. At larger effect sizes ($\delta \geq 0.25$), detection increased rapidly and approached saturation at relatively small sample sizes, particularly at ABBY.

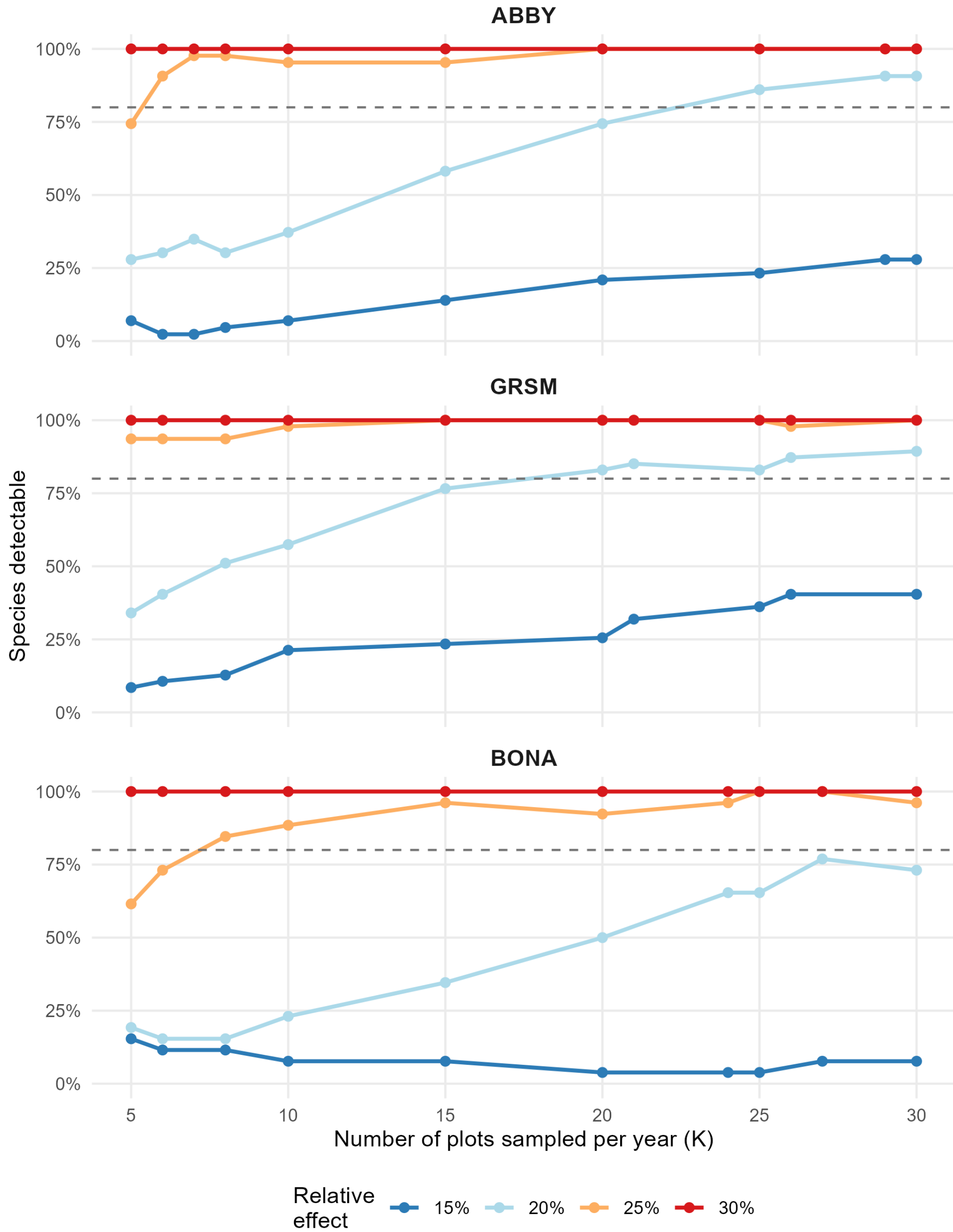

Figure 2: Counterfactual change detection across three example sites. Curves show the percentage of species with detection probability at least 0.80 as a function of the number of plots sampled per year (K), for imposed relative changes of 15–30%.

Full site-level detection curves for all sites are provided in Appendix A (Figures A2–A3), illustrating how detection responses vary across the Observatory beyond the representative examples shown here.

4.1.2 Summary across sites

Across sites, the number of plots required to achieve the detection criterion (K^*, the smallest K at which the detection threshold is met) varied substantially (Table 2). At $\delta = 0.20$, most sites reached the 80% detection threshold between approximately $K = 20$ and $K = 30$, although several sites did not reach the threshold within the evaluated range.

Detection increased monotonically with sample size at nearly all sites, but the rate of increase differed. At the maximum evaluated sample size ($K = 30$), detection also varied among sites, indicating that for some communities the 80% detection threshold was not achievable within the range of sampling effort considered. These differences reflect variation in community composition and spatial heterogeneity rather than differences in model behavior. Note that sites with very few species above the 3% cover floor (e.g., BART with 7, MOAB with 4, JORN with 5) can show low K^* values that reflect small denominators in the community-level metric rather than inherently easy detectability; at these sites, detecting change in a single additional species can shift the community percentage above the 80% threshold.

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Table 2: Summary of counterfactual change detection across sites. K^* is the smallest number of plots at which at least 80% of species exceeded the detection criterion for a 20% relative change. Detection at maximum K reports the percentage of species detectable at the highest available sample size for each site.

Domain	Site Code	Species (≥3% cover)	K*	Detection at K = 30 (%)
D01	BART	7	5	57.1
D03	JERC	122	10	94.3
D02	SERC	35	10	77.1
D14	SRER	38	10	89.5
D08	TALL	64	10	93.8
D11	CLBJ	63	15	95.2
D08	LENO	33	15	90.9
D13	MOAB	4	15	100.0
D17	SOAP	46	15	91.3
D05	STEI	71	15	88.7
D05	UNDE	44	15	90.9
D03	DSNY	92	20	91.3
D07	GRSM	47	20	89.4
D14	JORN	5	20	80.0

Domain	Site Code	Species (≥3% cover)	K*	Detection at K = 30 (%)
D11	OAES	47	20	87.2
D15	ONAQ	9	20	88.9
D07	ORNL	64	20	92.2
D03	OSBS	54	20	94.4
D16	ABBY	43	25	90.7
D01	HARV	47	25	83.0
D06	KONZ	52	25	88.5
D13	NIWO	30	25	86.7
D09	NOGP	49	25	83.7
D09	WOOD	45	25	86.7
D10	RMNP	19	30	84.2
D02	SCBI	58	30	81.0
D17	SJER	40	30	90.0
D06	UKFS	68	30	82.4
D16	WREF	23	30	82.6
D18	BARR	10	NA	70.0
D02	BLAN	30	NA	NA
D19	BONA	26	NA	73.1
D10	CPER	22	NA	68.2
D09	DCFS	36	NA	69.4
D19	DEJU	24	NA	70.8
D04	GUAN	77	NA	NA
D19	HEAL	36	NA	66.7
D06	KONA	9	NA	66.7
D04	LAJA	83	NA	63.9
D07	MLBS	48	NA	77.1
D20	PUUM	20	NA	NA
D10	STER	16	NA	NA
D17	TEAK	29	NA	NA
D18	TOOL	19	NA	47.4
D05	TREE	49	NA	NA
D12	YELL	29	NA	NA

\endgroup

Complete site-level detection results are provided in Appendix A (Table A1).

4.2 Multi-year trend detection

4.2.1 Example sites

We illustrate the behavior of the trend analysis using the same three representative sites. Trend detection differs from year-to-year change detection due to the combined influence of spatial and temporal variability in site-level means.

4.2.2 Detection and stability

Observed trend detectability was low across all three sites. At the maximum evaluated sample size, no site met the detection criterion for baseline trends (Table 3), indicating that directional changes present in the data could not be reliably distinguished from background variability at the site level.

For imposed trends, detection was highest at small sample sizes ($K \approx 5$) and declined with the inclusion of additional plots (Figure 3). Detection decreased rapidly through approximately $K = 10$ and then stabilized at roughly 50–60% of species across sites. At no site did detection reach the 80% threshold within the evaluated range of sample sizes. This pattern reflects spatial heterogeneity within each site across which plots are distributed. As additional plots are included, variability in annual site means increases, reducing the ability to distinguish directional change from background variation. The pattern is consistent across all sites (Appendix B2).

Full site-level trend detection and stability curves are provided in Appendix B (Figures B2–B3), demonstrating the consistency of these patterns across sites.

4.2.3 Summary across sites

Across the Observatory, sample size requirements for trend detection and stability varied among sites (Table 3). In most cases, the number of plots required to detect a 20% cumulative trend (Trend K^*) was smaller than the number required to achieve stable slope estimates (K_{CV50}), indicating that reliable estimation of trend magnitude requires more intensive sampling than detection alone.

At some sites, the detection threshold was not achieved within the evaluated range of sample sizes, even though slope estimates became progressively more stable with increasing K. Conversely, sites with lower variability showed similar sample size requirements for detection and stability. In many cases, Trend K* was not defined because the detection threshold was not reached within the evaluated range of sample sizes.

These patterns are synthesized with the year-to-year change detection results in Section 4.3.

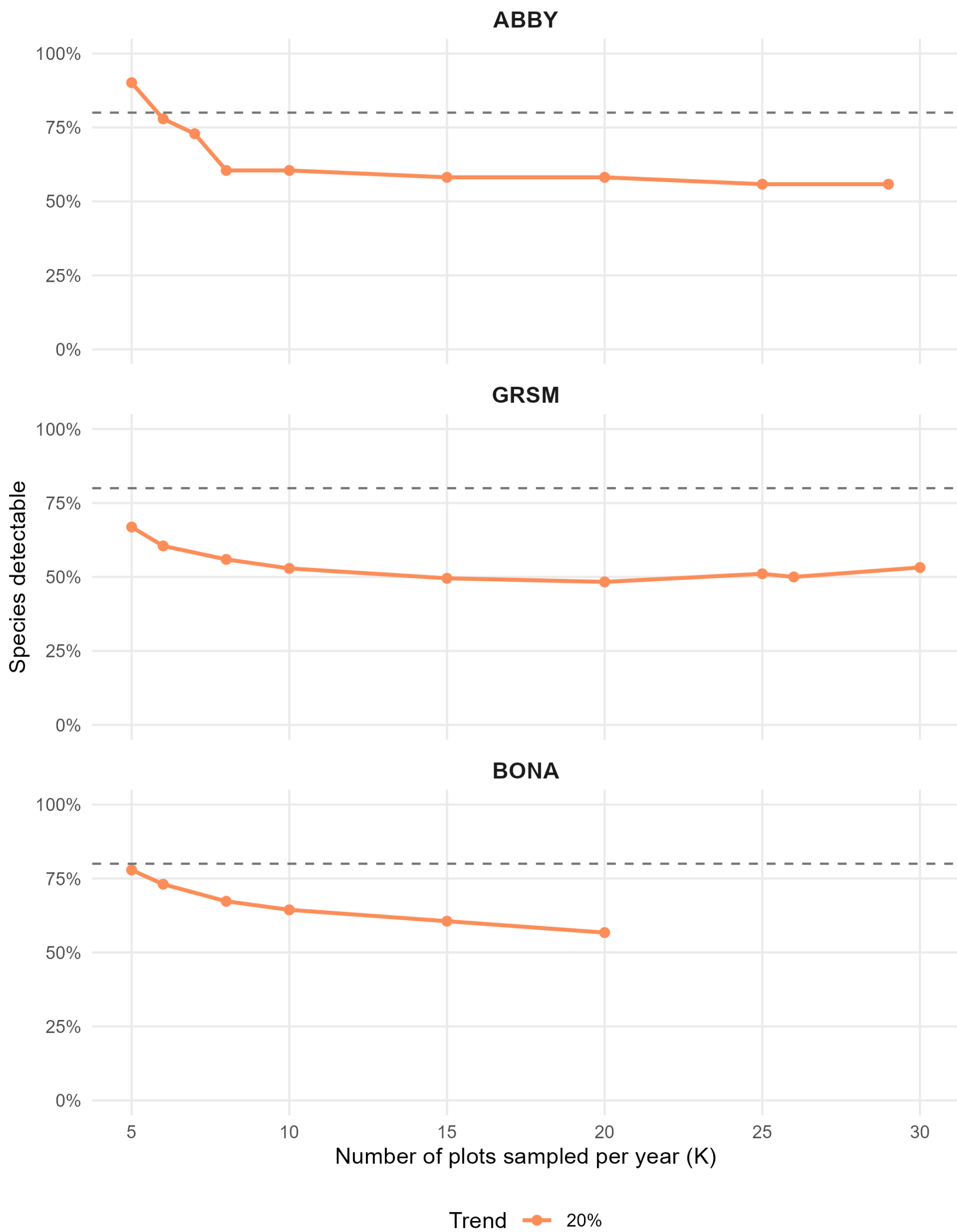

Figure 3: Trend detection across three example sites. Curves show the fraction of species with detection probability at least 0.80 as a function of the number of plots sampled per year (K), for imposed cumulative trends of different magnitudes.

Table 3: Summary of multi-year trend detection and slope stability across sites. Trend K* is the smallest number of plots at which at least 80% of species met the detection criterion for a 20% imposed cumulative trend. Detection at maximum K reports the percentage of species detectable at the highest available sample size. K_CV50 is the smallest number of plots at which the median slope CV was ≤ 0.50, and CV at maximum K reports the corresponding value at the highest available sample size.

Domain	Site Code	Trend K*	Detection at max K (%)	K_CV50	Slope CV at max K	Max K
D01	BART	NA	57.1	10	0.000	30
D01	HARV	NA	38.3	20	0.000	30
D02	BLAN	5	83.3	10	0.000	20
D02	SCBI	NA	58.6	15	0.160	24
D02	SERC	NA	49.1	20	0.000	30
D03	DSNY	NA	50.0	8	0.000	30
D03	JERC	NA	53.3	10	0.160	26
D03	OSBS	NA	59.3	5	0.093	20
D04	GUAN	NA	41.0	5	0.034	17
D04	LAJA	5	91.6	10	0.068	27
D05	STEI	NA	39.4	6	0.091	29
D05	TREE	5	63.3	6	0.000	26
D05	UNDE	NA	51.1	7	0.037	29
D06	KONA	NA	66.7	8	0.220	20
D06	KONZ	NA	46.2	13	0.061	29
D06	UKFS	5	60.3	15	0.250	16
D07	GRSM	NA	53.2	10	0.000	30
D07	MLBS	5	71.4	5	0.090	12
D07	ORNL	NA	53.1	15	0.140	25
D08	LENO	NA	51.5	5	0.026	29
D08	TALL	NA	49.0	10	0.061	29
D09	DCFS	5	55.6	5	0.000	30
D09	NOGP	NA	49.0	5	0.000	30
D09	WOOD	NA	55.6	7	0.160	26
D10	CPER	NA	72.7	5	0.015	29
D10	RMNP	NA	68.4	5	0.000	30
D10	STER	NA	65.3	5	0.000	7
D11	CLBJ	NA	70.9	5	0.350	7
D11	OAES	NA	62.8	8	0.000	30
D12	YELL	NA	72.4	5	0.340	7

Domain	Site Code	Trend K*	Detection at max K (%)	K_CV50	Slope CV at max K	Max K
D13	MOAB	5	75.0	6	0.130	15
D13	NIWO	NA	50.0	20	0.000	30
D14	JORN	NA	60.0	5	0.000	30
D14	SRER	NA	47.9	5	0.029	27
D15	ONAQ	NA	72.2	5	0.190	17
D16	ABBY	5	55.8	7	0.063	29
D16	WREF	NA	76.8	5	0.100	7
D17	SJER	5	70.6	5	0.240	10
D17	SOAP	NA	54.3	5	0.160	19
D17	TEAK	NA	31.0	10	0.180	17
D18	BARR	NA	73.3	5	0.160	19
D18	TOOL	NA	63.2	10	0.410	11
D19	BONA	NA	56.7	20	0.380	20
D19	DEJU	5	63.3	20	0.230	24
D19	HEAL	5	89.4	NA	0.800	8
D20	PUUM	NA	50.0	12	0.000	20

4.3 Synthesis across analyses

The year-to-year change detection and trend analyses provide complementary but distinct perspectives on the detectability of ecological change. Both analyses show that increasing the number of sampled plots improves inference, but they differ in what constitutes sufficient sampling.

The year-to-year change detection analysis evaluates the ability to detect a discrete change, and indicates that the detection criterion for a 20% relative change is often met at sample sizes on the order of $K \approx 20$–25. In contrast, the trend analysis shows that reliable estimation of directional change imposes a more stringent requirement. Trend detection thresholds were generally not achieved within the evaluated range of sample sizes, even when detection appeared relatively high at small K, whereas stability improved consistently with increasing K. Full site-level detection and stability curves are provided in Appendices A and B, allowing direct comparison of how inference responds to sampling effort across all sites.

Where both approaches converge on similar sample size requirements, detection and estimation are aligned. Where they diverge, detection alone can overstate the reliability of inference, as slope estimates may remain sensitive to plot selection even when detectability appears relatively high. This distinction reflects the difference between identifying change and quantifying its magnitude.

Together, these results show that detectability and stability represent distinct constraints on inference. Detection determines whether change can be identified, whereas stability determines whether that change can be estimated reliably. Reliable ecological inference therefore requires sufficient sampling not only to detect change, but to ensure that estimates of that change are robust to spatial variability.

Full site-level detection and stability curves are provided in Appendices A and B to show variation in curve shape across the Observatory beyond the summary values reported in Tables 2 and 3.

4.4 Recommendation

To synthesize the results of the year-to-year change detection and trend analyses into site-level guidance, we defined a recommended sample size (K_{rec}) that integrates both detection and stability constraints. For each site, K_{rec} was defined as the maximum of the sample size required to achieve the detection criterion for a 20% change (Tier0 K^*), the sample size required to achieve stable trend estimation (K_{CV50}), and a minimum threshold of $K = 15$ plots to avoid undersampling in spatially heterogeneous communities.

The minimum threshold of $K = 15$ reflects the consistent pattern observed across sites that slope stability improves rapidly between approximately 5 and 15 plots, with diminishing gains beyond this range (Table 3, Figure 7). Of the 46 sites analyzed, all but one achieved K_{CV50} at or below $K = 20$, and the majority reached this threshold at $K \leq 10$. However, sampling below 15 plots frequently resulted in slope estimates that were highly sensitive to plot selection, even when detection metrics appeared favorable. This safeguard ensures that recommended sampling designs support not only the detection of ecological change, but also the reliable estimation of its magnitude.

Trend detection thresholds were not used directly in defining recommended sample sizes, because detectability of imposed trends decreased with increasing sample size and did not provide a consistent basis for design decisions (Appendix B2). Instead, slope stability provides a more robust and interpretable criterion for determining the sampling effort required for reliable trend inference.

Together, the appendix results illustrate that the two analyses provide complementary constraints on sampling design. Year-to-year change detection defines the minimum sampling required to reliably identify change, while trend stability defines the sampling required to ensure that estimates of change are robust to spatial variability. Where these constraints converge, recommended sample sizes are strongly supported. Where they diverge, the more stringent requirement governs the recommendation.

The appendix figures provide the underlying detection and stability curves used to derive these recommendations, allowing direct evaluation of how each site-level recommendation reflects the full response of inference quality to sampling effort (Appendices A and B).

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Table 4: Site-level recommended sample sizes integrating year-to-year change detection and trend stability. YY K* is the smallest number of plots at which the year-to-year detection criterion was met. K_CV50 is the smallest number of plots at which median slope CV was ≤ 0.50. K rec is the maximum of YY K, *K_CV50, and a minimum safeguard of K = 15; when YY K* was not reached, the recommendation is to retain the maximum available K.

Domain	Site Code	YY K*	K_CV50	Max K	K rec	Basis
D01	BART	5	10	30	15	Minimum design safeguard
D01	HARV	25	20	30	25	Year-to-year detection constraint
D02	BLAN	NA	10	20	20	Retain max K: year-to-year detection threshold not reached
D02	SCBI	30	15	24	30	Year-to-year detection constraint
D02	SERC	10	20	30	20	Trend stability constraint
D03	DSNY	20	8	30	20	Year-to-year detection constraint
D03	JERC	10	10	26	15	Minimum design safeguard
D03	OSBS	20	5	20	20	Year-to-year detection constraint
D04	GUAN	NA	5	17	17	Retain max K: year-to-year detection threshold not reached
D04	LAJA	NA	10	27	27	Retain max K: year-to-year detection threshold not reached
D05	STEI	15	6	29	15	Year-to-year detection constraint
D05	TREE	NA	6	26	26	Retain max K: year-to-year detection threshold not reached
D05	UNDE	15	7	29	15	Year-to-year detection constraint
D06	KONA	NA	8	20	20	Retain max K: year-to-year detection threshold not reached
D06	KONZ	25	13	29	25	Year-to-year detection constraint
D06	UKFS	30	15	16	30	Year-to-year detection constraint
D07	GRSM	20	10	30	20	Year-to-year detection constraint
D07	MLBS	NA	5	12	12	Retain max K: year-to-year detection threshold not reached
D07	ORNL	20	15	25	20	Year-to-year detection constraint
D08	LENO	15	5	29	15	Year-to-year detection constraint
D08	TALL	10	10	29	15	Minimum design safeguard
D09	DCFS	NA	5	30	30	Retain max K: year-to-year detection threshold not reached
D09	NOGP	25	5	30	25	Year-to-year detection constraint
D09	WOOD	25	7	26	25	Year-to-year detection constraint

Domain	Site Code	YY K*	K_CV50	Max K	K rec	Basis
D10	CPER	NA	5	29	29	Retain max K: year-to-year detection threshold not reached
D10	RMNP	30	5	30	30	Year-to-year detection constraint
D10	STER	NA	5	7	7	Retain max K: year-to-year detection threshold not reached
D11	CLBJ	15	5	7	15	Year-to-year detection constraint
D11	OAES	20	8	30	20	Year-to-year detection constraint
D12	YELL	NA	5	7	7	Retain max K: year-to-year detection threshold not reached
D13	MOAB	15	6	15	15	Year-to-year detection constraint
D13	NIWO	25	20	30	25	Year-to-year detection constraint
D14	JORN	20	5	30	20	Year-to-year detection constraint
D14	SRER	10	5	27	15	Minimum design safeguard
D15	ONAQ	20	5	17	20	Year-to-year detection constraint
D16	ABBY	25	7	29	25	Year-to-year detection constraint
D16	WREF	30	5	7	30	Year-to-year detection constraint
D17	SJER	30	5	10	30	Year-to-year detection constraint
D17	SOAP	15	5	19	15	Year-to-year detection constraint
D17	TEAK	NA	10	17	17	Retain max K: year-to-year detection threshold not reached
D18	BARR	NA	5	19	19	Retain max K: year-to-year detection threshold not reached
D18	TOOL	NA	10	11	11	Retain max K: year-to-year detection threshold not reached
D19	BONA	NA	20	20	20	Retain max K: year-to-year detection threshold not reached
D19	DEJU	NA	20	24	24	Retain max K: year-to-year detection threshold not reached
D19	HEAL	NA	NA	8	8	Retain max K: year-to-year detection threshold not reached
D20	PUUM	NA	12	20	20	Retain max K: year-to-year detection threshold not reached

```
## \endgroup
```

5 DISCUSSION

Monitoring design is constrained not only by how easily ecological change can be detected, but by how it is defined and quantified. By integrating year-to-year change detection and multi-year trend analyses, we derived site-level recommendations for spatial replication that explicitly balance the ability to detect change with the reliability of estimated trends.

Across NEON plant communities, sample sizes sufficient for detecting discrete change are often insufficient for reliably estimating multi-year trends, and effective monitoring design must satisfy both constraints. This distinction between detecting change and estimating its magnitude is well established in statistical inference and ecological modeling (Clark et al. 2017; Gelman et al. 2013).

5.1 Complementary constraints on inference

This analysis shows that conclusions about required spatial replication - number of plots across heterogeneous sites - depend not only on the magnitude of ecological change, but also on how inference is defined. Required sample size therefore depends on both ecological variability and the statistical definition of inference (Gotelli and Ellison 2013). The year-to-year change detection and multi-year trend analyses provide complementary perspectives on this question.

The year-to-year change detection analysis evaluates the ability to detect a specified relative change at a single time point, providing guidance relevant to monitoring abrupt ecological shifts. The trend analysis evaluates the reliability of estimated rates of change over multi-year windows, providing guidance relevant to gradual directional processes.

Across the Observatory, both analyses point to a similar range of sampling effort, with most sites requiring approximately 20–25 plots to achieve community-level detection of a 20% change and to produce stable trend estimates (median CV ≤ 0.50). The convergence of these independent analyses, which operate on different temporal scales and statistical criteria, strengthens confidence in this range.

At the same time, the analyses highlight that detection and estimation are not equivalent. Detection thresholds were often reached at smaller sample sizes than those required for stable trend estimation. In many cases, slope estimates remained sensitive to which plots were sampled even after detection thresholds were met. This indicates that detection alone can overstate the reliability of inference, and that stability provides a necessary complement for evaluating monitoring design. These distinctions are consistent with broader challenges in ecological monitoring, where detecting change and estimating trends often require different sampling strategies (Urquhart et al. 1998; National Research Council 2001).

5.2 Stability as a design criterion

The slope stability framework provides a direct and interpretable measure of inference quality. The coefficient of variation of slope estimates quantifies the sensitivity of trend estimates to spatial sampling, and decreases monotonically with increasing sample size at nearly all sites.

The CV threshold (CV ≤ 0.50) defines an analogue to the K^* concept used in detection-based analyses: it identifies the point at which additional sampling no longer qualitatively changes the inferred trend. This provides a practical and interpretable criterion for monitoring design. Unlike detection thresholds, which depend on an arbitrary choice of effect size, the stability criterion is intrinsic to the data and reflects the spatial heterogeneity of the system. Spatial heterogeneity is a fundamental driver of ecological variance and scaling relationships across systems (Levin 1992; Legendre and Legendre 2012).

The contrast between detection and stability is particularly relevant to trend inference. Detection-based trend metrics can exhibit non-monotonic behavior as additional plots are added, reflecting increased spatial variability in annual site means. This spatial dilution effect is ecologically real, but complicates interpretation of detection curves. In contrast, the stability metric directly evaluates the precision of the slope estimate and provides a consistent basis for comparing sites.

5.3 Per-species scaling and the role of filtering

The species specific perturbation and detection thresholds scaled to mean cover values from the field data ensure that all species face equivalent proportional challenges. This avoids the bias inherent in fixed absolute thresholds, where a given percentage-point change has very different ecological meaning across the abundance spectrum.

The 3% cover minimum threshold defines the portion of the community for which meaningful inference is possible. Below this threshold, a 20% relative change falls within the resolution of the field observations and cannot be reliably distinguished from measurement noise. As a result, the analyses characterize detection and stability for the more abundant component of the community, where ecological change can be resolved with confidence. Such constraints are consistent with well-documented patterns of species rarity and detection limits in ecological communities (Gaston 1994).

At most NEON sites, a large fraction of species fall below this threshold (Table 1). This does not represent a limitation of the modeling framework, but rather a constraint imposed by the combination of ecological rarity and measurement precision. The filtering step therefore clarifies the scope of inference rather than restricting it.

5.4 Modeling framework and dimension reduction

The use of GJAM enables joint inference across species, accounting for shared environmental responses and residual covariance. This extends the univariate framework used in the 2018 optimization analysis (Barnett et al. 2018), which could not capture coordinated responses among species. However, the joint modeling framework introduces compositional constraints and requires dimension reduction at NEON scales, and the single-fit design means that the gains from multivariate structure are partially offset by the optimistic sample size estimates discussed in Section 5.7.

Dimension reduction (Taylor-Rodriguez et al. 2017) is essential for applying this framework at NEON scales. At most sites, the number of species exceeds the effective sample size required to estimate a full covariance matrix. The reduced-rank representation captures dominant modes of association while maintaining computational tractability, allowing the model to scale across the full diversity gradient of the Observatory.

5.5 Implications for NEON sampling design

The results indicate that NEON's current design of approximately 20–30 plots per site is well-calibrated for detecting 20% relative changes in plant cover and for producing stable trend estimates over multi-year windows. This conclusion is supported by both the perturbation-based detection analysis and the trend stability analysis.

Changes smaller than approximately 15% are not reliably detectable under observed spatial variability at any evaluated sample size, establishing a practical lower bound on sensitivity. Conversely, changes of 25% or greater are detectable at relatively small sample sizes, indicating that large ecological shifts will be captured regardless of sampling intensity.

Site-level variation in required sample size reflects underlying spatial heterogeneity. Homogeneous systems can support reliable inference with relatively few plots, whereas more complex systems require greater spatial replication. The framework presented here provides a quantitative basis for evaluating these differences and adapting sampling strategies accordingly. The appendix figures show that while the general patterns described here are consistent, the shape and rate of change of detection and stability curves vary substantially among sites (Appendices A and B). For sites where the year-to-year detection threshold was not reached within the evaluated range (Table 4, "retain max K" sites), the recommendation to maintain current sampling levels is conservative by design. At some of these sites, alternative analytical approaches—such as functional group aggregation or ordination-based community metrics—may provide additional sensitivity to change that the per-species framework does not capture.

These analyses are intended to identify where sampling effort can be reduced without compromising NEON's ability to detect and estimate ecological change. At a small number of sites, the recommended sample size meets or exceeds the maximum number of plots currently available (e.g., UKFS, WREF, SJER), suggesting that current sampling intensity may be at or below the level needed for robust per-species inference. Increasing plot counts at these sites is not feasible under current funding constraints, and these results should not be interpreted as a recommendation to do so. The data collected at these sites remain valuable: they support joint species modeling, contribute to multi-site comparisons and Observatory-wide syntheses, and provide baseline records against which future change can be assessed as additional years of observation accumulate.

5.6 Addressing the 2018 trend detection question

The 2018 design optimization effort identified trend detection as a priority question but could not address it due to limited temporal replication (Barnett et al. 2018). With 8–12 years of data now available, the moving-window trend analysis presented here directly addresses that gap.

The key result is that reliable trend estimation imposes similar sample size requirements to those identified for detecting discrete changes. The stability framework provides an actionable interpretation of this result, linking sample size directly to the reliability of estimated rates of change. Together with the year-to-year change detection results, these findings provide the quantitative basis for resuming plant diversity sampling at site-specific plot counts as specified in Table 4.

5.7 Limitations and future directions

Several limitations warrant consideration. First, the analysis is based on a single model fit per site, and does not propagate uncertainty associated with refitting the model under different plot subsets. As a result, the reported sample size requirements represent optimistic estimates of detection and stability, as they do not propagate uncertainty associated with model refitting across spatial subsamples (Gelman et al. 2013). A refit-based analysis, in which the model is re-estimated for each subsampled dataset, would propagate uncertainty associated with parameter estimation and is expected to increase the sample sizes required for both detection and stability. This represents an important direction for future work and would provide a more conservative assessment of monitoring requirements.

Second, the analysis focuses on a range of relative effect sizes (15–30%), with 20% representing a critical threshold for detectability. Sensitivity to alternative detection thresholds was not explored and could be evaluated in future work.

Third, the compositional structure of the GJAM model compresses absolute cover values relative to raw field observations. This motivates the use of raw field cover for defining perturbations and detection thresholds, but introduces a layer of abstraction that should be considered when interpreting results.

The framework developed here clarifies that sampling adequacy depends on how ecological change is defined. This dependence reflects a broader distinction between hypothesis testing and parameter estimation in statistical inference (Gelman et al. 2013). Sample sizes sufficient for detecting discrete change are not necessarily sufficient for reliably estimating rates of change over time. Across NEON plant communities, both analyses indicate that approximately 20–25 plots are required for robust inference, but for different reasons: detection thresholds are met at this scale, and slope estimates become stable with respect to spatial sampling.

This distinction has direct implications for monitoring design. Designs based solely on detection criteria may underestimate the sampling effort required for reliable ecological inference. Incorporating stability-based metrics provides a more complete basis for evaluating monitoring performance, particularly for questions involving gradual, multi-year change. As NEON and similar observatories continue to mature, distinguishing between detectability and reliability will be essential for ensuring that monitoring design supports not just the detection of change, but its reliable interpretation.

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A Appendix A. Year-to-year change detection

A.1 Summary table

Table 5: Full summary of year-to-year change detection across all sites.

Domain	Site Code	K*	Detection at K=30 (%)	Spp. (u22653%)
D01	BART	5	57.1	7
D01	HARV	25	83.0	47
D02	BLAN	NA	NA	30
D02	SCBI	30	81.0	58
D02	SERC	10	77.1	35
D03	DSNY	20	91.3	92
D03	JERC	10	94.3	122
D03	OSBS	20	94.4	54
D04	GUAN	NA	NA	77
D04	LAJA	NA	63.9	83
D05	STEI	15	88.7	71
D05	TREE	NA	NA	49
D05	UNDE	15	90.9	44
D06	KONA	NA	66.7	9
D06	KONZ	25	88.5	52
D06	UKFS	30	82.4	68
D07	GRSM	20	89.4	47
D07	MLBS	NA	77.1	48
D07	ORNL	20	92.2	64
D08	LENO	15	90.9	33
D08	TALL	10	93.8	64
D09	DCFS	NA	69.4	36
D09	NOGP	25	83.7	49
D09	WOOD	25	86.7	45
D10	CPER	NA	68.2	22
D10	RMNP	30	84.2	19
D10	STER	NA	NA	16

Domain	Site Code	K*	Detection at K=30 (%)	Spp. (u22653%)
D11	CLBJ	15	95.2	63
D11	OAES	20	87.2	47
D12	YELL	NA	NA	29
D13	MOAB	15	100.0	4
D13	NIWO	25	86.7	30
D14	JORN	20	80.0	5
D14	SRER	10	89.5	38
D15	ONAQ	20	88.9	9
D16	ABBY	25	90.7	43
D16	WREF	30	82.6	23
D17	SJER	30	90.0	40
D17	SOAP	15	91.3	46
D17	TEAK	NA	NA	29
D18	BARR	NA	70.0	10
D18	TOOL	NA	47.4	19
D19	BONA	NA	73.1	26
D19	DEJU	NA	70.8	24
D19	HEAL	NA	66.7	36
D20	PUUM	NA	NA	20

A.2 Detection curves (20% change)

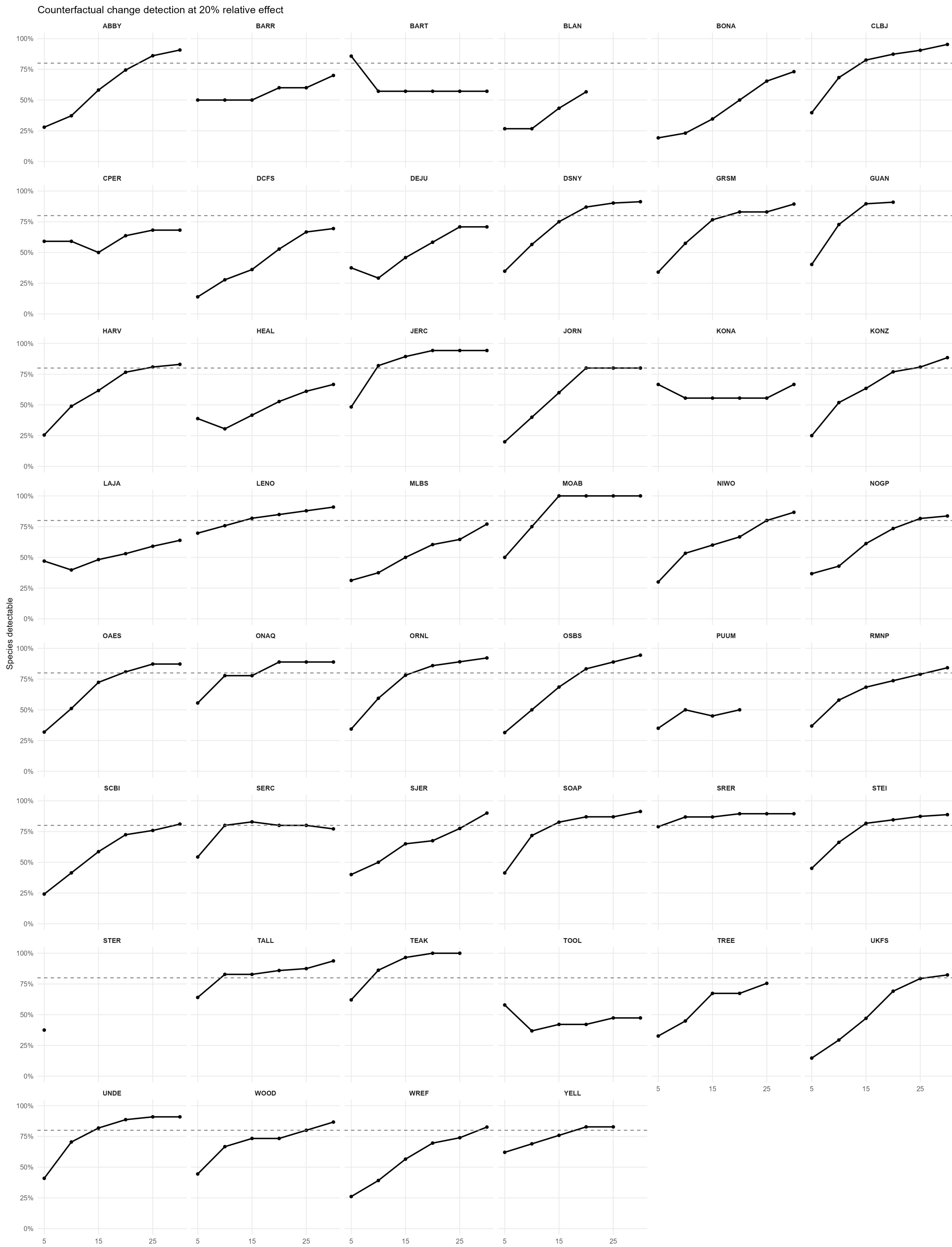

Figure 4: Year-to-year change detection curves for all sites at a 20% relative change.

A.3 Detection curves (all effect sizes)

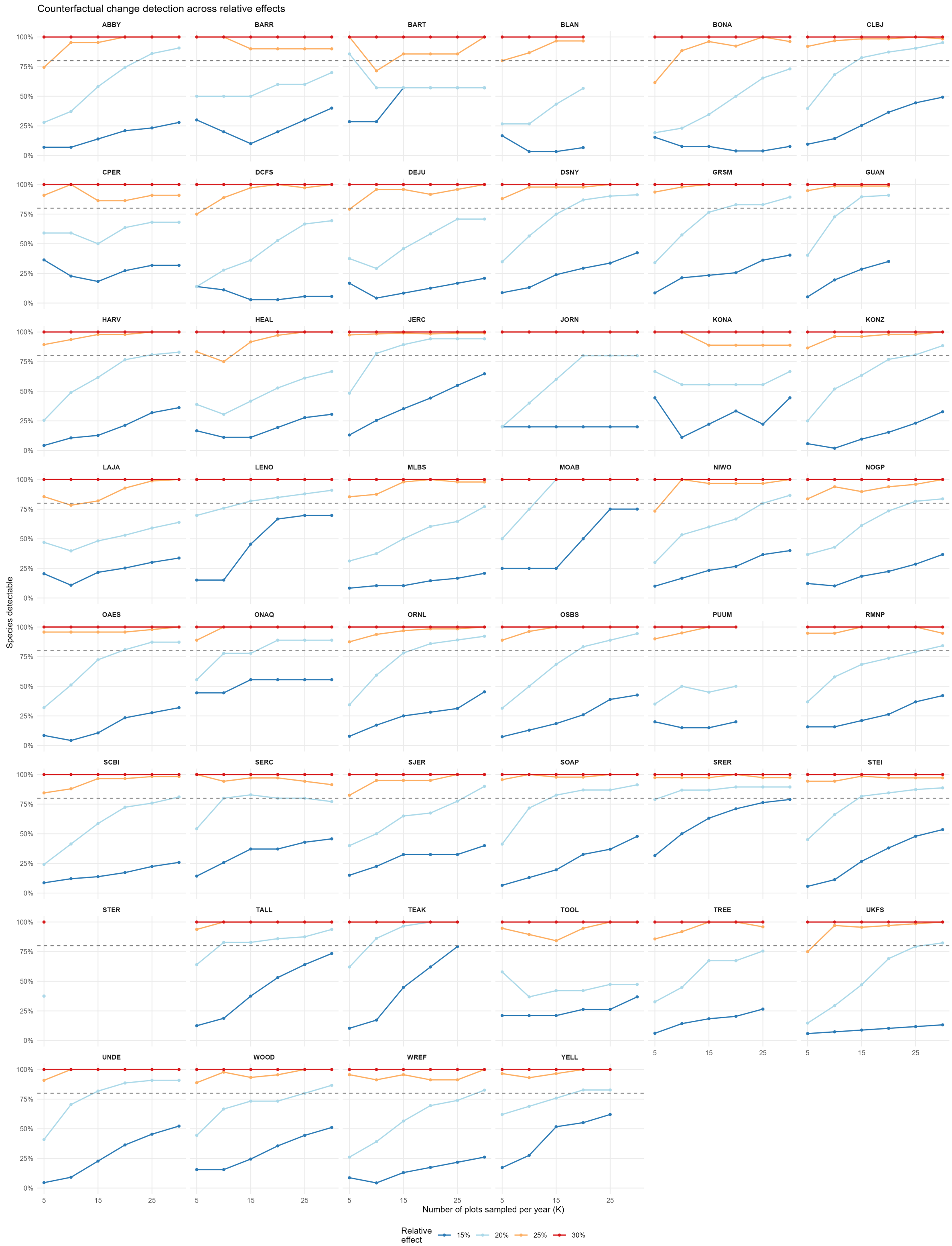

Figure 5: Year-to-year change detection curves for all sites across evaluated effect sizes (15–30%).

B Appendix B. Multi-year trend analysis

B.1 Summary table

Table 6: Full summary of multi-year trend detection and slope stability across all sites.

Domain	Site Code	Species (≥3% cover)	trend K*	trend Detection at K=30 (%)	K_CV50	CV at K=30
D01	BART	7	NA	57.1	10	0
D01	HARV	47	NA	38.3	20	0
D02	BLAN	30	5	NA	10	NA
D02	SCBI	58	NA	NA	15	NA
D02	SERC	35	NA	49.1	20	0
D03	DSNY	92	NA	50.0	8	0
D03	JERC	122	NA	NA	10	NA
D03	OSBS	54	NA	NA	5	NA
D04	GUAN	77	NA	NA	5	NA
D04	LAJA	83	5	NA	10	NA
D05	STEI	71	NA	NA	6	NA
D05	TREE	49	5	NA	6	NA
D05	UNDE	44	NA	NA	7	NA
D06	KONA	9	NA	NA	8	NA
D06	KONZ	52	NA	NA	13	NA
D06	UKFS	68	5	NA	15	NA
D07	GRSM	47	NA	53.2	10	0
D07	MLBS	48	5	NA	5	NA
D07	ORNL	64	NA	NA	15	NA
D08	LENO	33	NA	NA	5	NA
D08	TALL	64	NA	NA	10	NA
D09	DCFS	36	5	55.6	5	0
D09	NOGP	49	NA	49.0	5	0
D09	WOOD	45	NA	NA	7	NA
D10	CPER	22	NA	NA	5	NA
D10	RMNP	19	NA	68.4	5	0
D10	STER	16	NA	NA	5	NA
D11	CLBJ	63	NA	NA	5	NA
D11	OAES	47	NA	62.8	8	0

Domain	Site Code	Species (≥3% cover)	trend K*	trend Detection at K=30 (%)	K_CV50	CV at K=30
D12	YELL	29	NA	NA	5	NA
D13	MOAB	4	5	NA	6	NA
D13	NIWO	30	NA	50.0	20	0
D14	JORN	5	NA	60.0	5	0
D14	SRER	38	NA	NA	5	NA
D15	ONAQ	9	NA	NA	5	NA
D16	ABBY	43	5	NA	7	NA
D16	WREF	23	NA	NA	5	NA
D17	SJER	40	5	NA	5	NA
D17	SOAP	46	NA	NA	5	NA
D17	TEAK	29	NA	NA	10	NA
D18	BARR	10	NA	NA	5	NA
D18	TOOL	19	NA	NA	10	NA
D19	BONA	26	NA	NA	20	NA
D19	DEJU	24	5	NA	20	NA
D19	HEAL	36	5	NA	NA	NA
D20	PUUM	20	NA	NA	12	NA

B.2 Trend detection curves (20% cumulative change)

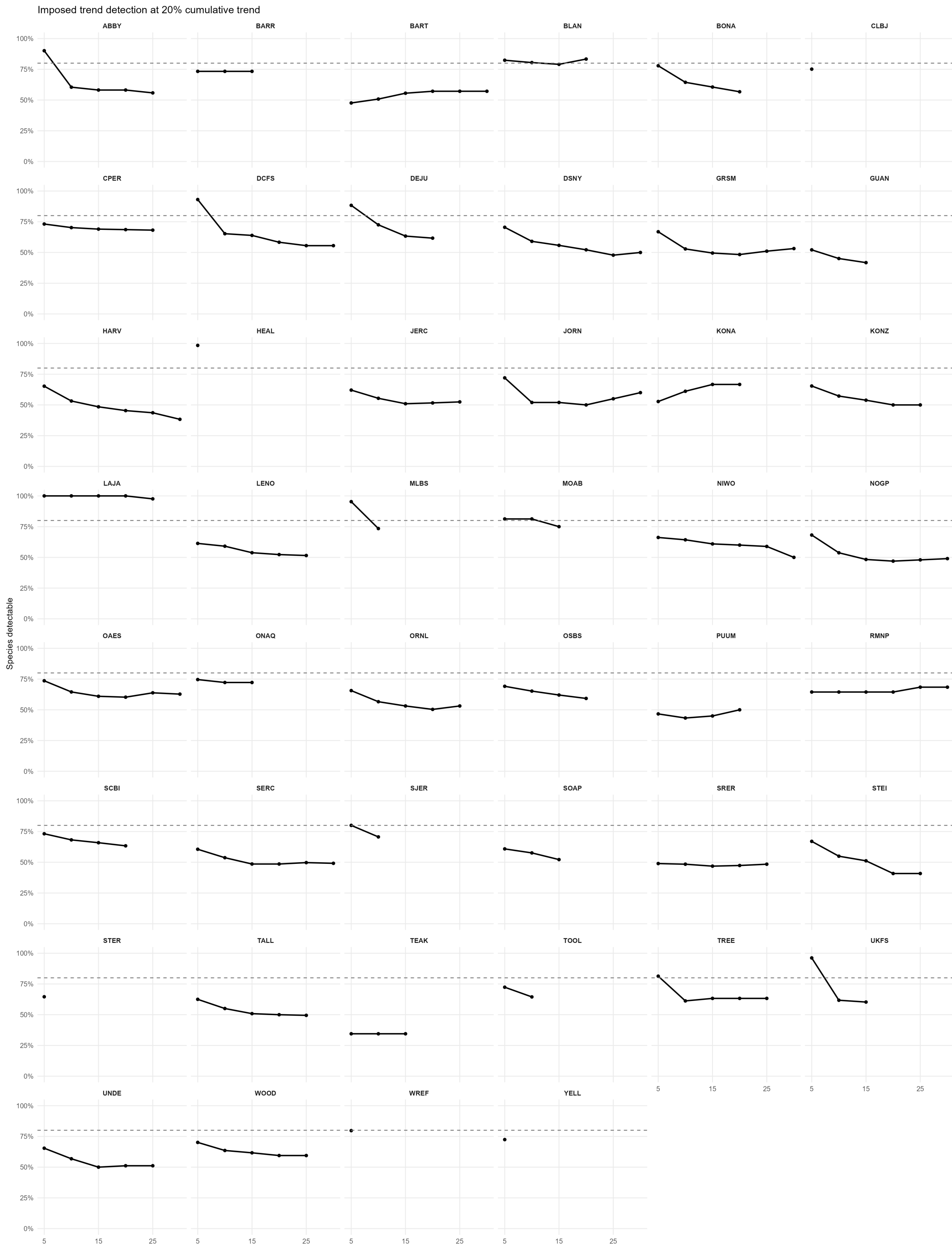

Figure 6: Trend detection curves for all sites under a 20% cumulative change scenario.

B.3 Slope stability

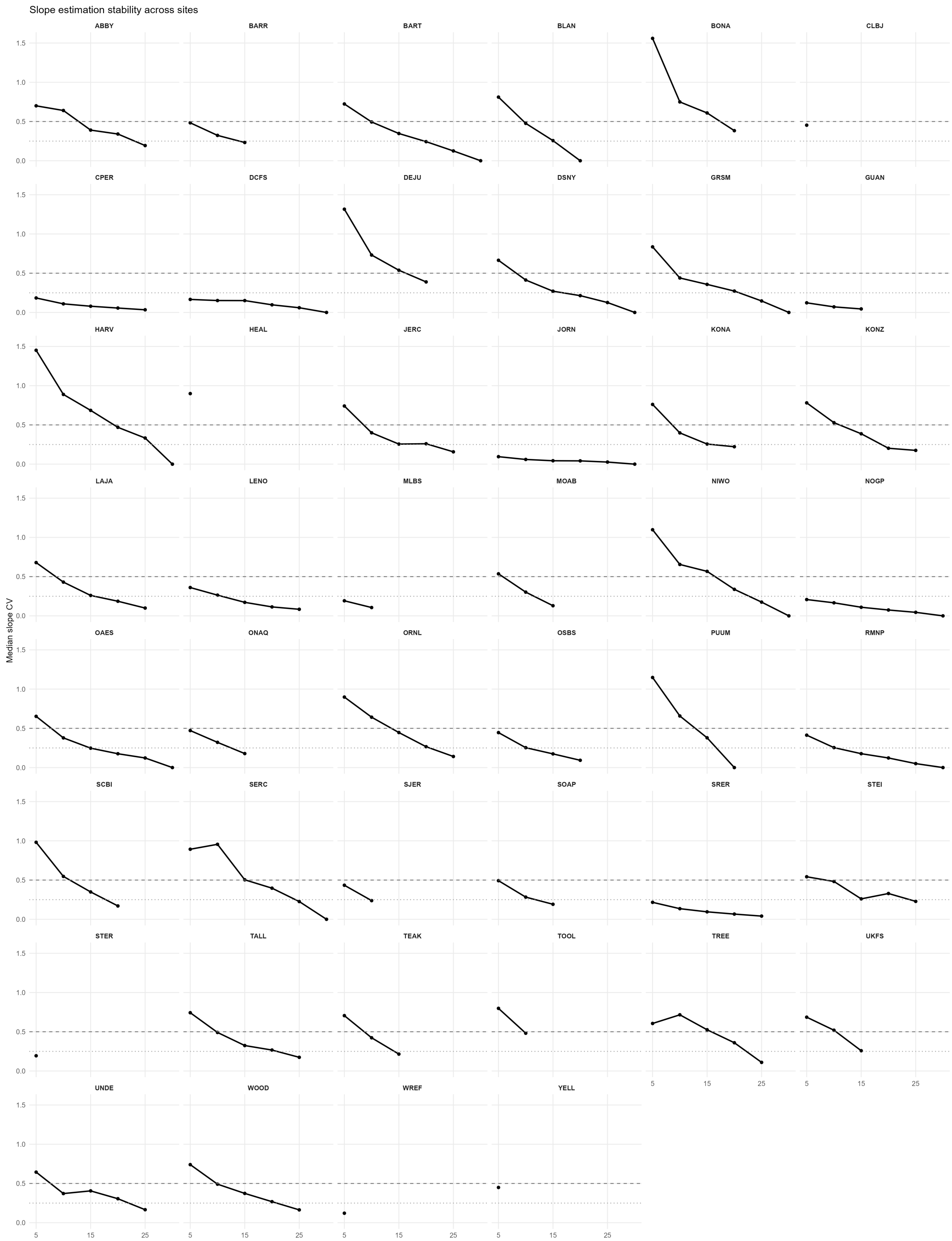

Figure 7: Slope stability across all sites, showing the coefficient of variation (CV) of slope estimates as a function of sample size (K).